

Project Repository Results Appendix

Are Natural Disasters Disastrous for Education? Evidence from Seven Asian Countries

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Alternative sample weighting schemes

In this website document, we provide additional summary statistics and regression results using alternative weighting schemes. See Section A.4 for more discussions on using MICS weights in our empirical setting. We generate the additional results separately using two types of sample weight variables. One is using directly *fsweight*. The other one is a variable generated with *fsweight* and additional information.

Modify *fsweight* with Region-Year Specific Population Size. Because this study is cross-region, the sample weight variable should account for the differences in population size across regions and years. Rather than only using *fsweight*, we propose a sample weight variable generated by ourselves, because of the concern on if *fsweight* is correctly capturing the difference in population size across region and year. Instead, the appropriate weight for each observation is defined as the product of the MICS-provided sampling weight and a “region weight”. The region weight is designed to be proportional to the number of children in the region and inversely proportional to the number of sampled children from that region.

Let $fsweight_{ic}$ denote the raw weight variable for child i in survey c . The value of $fsweight_{ic}$ indicates the number of children represented by that observation, accounting for the survey’s stratification criteria. For example, in the Bangladesh 2019 survey (Table C.13), the sum of *fsweight* across all children is 66,705, while the dataset contains 40,617 observations. This demonstrates that *fsweight* provides appropriate relative sampling weights within the region surveyed. Let N_c denote the number of observations in survey c , $ChildPop_c$ the size of child population in the relevant region and year, and m the total number of surveys. The final weight assigned to each observation is calculated as:

$$P_{ic} = \frac{fsweight_{ic}}{\sum_{i=1}^{N_c} fsweight_{ic}} \times \frac{ChildPop_c}{\sum_{c=1}^m ChildPop_c}$$

Table C.13 reports the values used in this formula for each region-year. Because population data by single-year age groups are not available, we approximate $ChildPop_c$ using the population of children aged 5-19 from <https://www.populationpyramid.net/>. For Pakistan, the raw weight variable *fsweight* is generated at the province-year level, so the ideal case would be that we obtain the province population for children in certain age range. However, we do not have direct measure of children population of each province. Hence, we calculate the children population size of each province as the population aged 5-19 in the country divided by the total population of the country, multiplied by the total population of the province. The population size of Khyber Pakhtunkhwa in 2017 is 35.5 million; Punjab in 2017 is 109.99 million; and Sindh in 2017 is 47.85 million. The population aged 5–19 for the entire country is 80.366 million, while

the total population of the country is 226.929 million. For example, the $ChildPop_c$ used for Khyber Pakhtunkhwa is $35.5/226.929 * 80.366 = 12.572$ million. For Thailand, the raw sample weight variable $fsweight$ is generated at the country-year level, so we directly use the population size for the whole country for two samples. Second, we present the number of observations in the raw datasets of children aged 5-17 for each country (region). Finally, we present the sum of the sample weight variable ($fsweight$) in each dataset. These are the value of $\sum_{i=1}^{N_c} fsweight_{ic}$.

Use $fsweight$ Directly. Although there are issues we do have answers for about the $fsweight$ in MICS, one may still want to apply the sample weight to summary statistics and analysis directly using this variable. In fact, according to Table C.13, one may notice that MICS generates the variable $fsweight$ in a way trying to capture the differences in population size across regions and years. For example, the sum of $fsweight$ in Kyrgyz Republic is similar to that in Turkmenistan, and their population size are indeed close. In the case where we believes that $fsweight$ fully captures the differences in population size across regions and years, we re-run the summary statistics and analysis using $fsweight$ directly, and find the results are quite consistent with those using the above variable.

Weighted and Unweighted Estimation Results Comparisons

In our multi-country setting, weights should capture cross country relative child population sizes. We collected best-approximate (see Appendix A.4) child population sizes corresponding to the MICS survey region/time/age-range, and our measures (column 3 in Table C.13) correlate with the sum of MICS weights (column 6 of Table C.13) for each survey region. This indicates to us that MICS weights do attempt to take cross-country population size into account, although we did not find clear documentations for how that is done. It is plausible that the MICS cross-country weights are computed given undocumented particularities of these surveys and capture better the cross-country representativeness. To be robust, we developed two alternative weighting schemes. In the first weighting scheme, results for which are shown in website-documents Tables D.1-D.16, we capture within-country relative sample weights using MICS weights and cross-country relative sample weights using our separately collected child population sizes information (column 3 in Table C.13). In the second weighting scheme, results for which are shown in website-documents Tables E.1-E.16, we directly use the MICS weights as provided for both within-country and cross-country representativeness.

We summarize here estimation results with and without weights included in the website-documents. We find a broadly similar set of findings across the two weighting schemes and our unweighted results (unchanged from the prior submission). When weights are used, there are reductions in the precision of estimates among most results perhaps due to a combination of the efficiency issue raised by Solon, Haider and Wooldridge (2015) as well as a loss in sample size. This leads to changes in which estimates exceed or fall below significance thresholds. Throughout our unweighted and two sets of weighted results, we find consistent evidence that early life disaster exposures negatively impact later enrollment and math test scores.

First, in terms of standard errors, the average enrollment disaster effects in the unweighted Table 5 have smaller standard errors than estimates from weighted Tables D.6 and

E.6. Comparing math score effects findings in Table 6 with those in Tables D.7 and E.7, standard errors increased for 14 and 15 out of 16 estimates, respectively.

Second, in terms of the point estimates of early life disaster enrollment effects, in Tables 5, D.6 and E.6, we find similar magnitudes and directions of average effects, and we lose statistical significance for the effects of all disasters in Table D.6. Considering heterogeneities by gender and age, the directions of early disaster enrollment estimates in Tables 7 and D.8 are identical for all 24 estimates, and identical for 23 out of 24 estimates in Tables 7 and E.8. For estimates of early life disaster math score effects, in Tables 6, D.7, and E.7, we find similar magnitudes and directions of average effects. For example, the effects of early life all disasters on math score are -0.031 (s.e. 0.015) in Table 6, -0.015 (s.e. 0.019) in Table D.7, and -0.027 (s.e. 0.018) in Table E.7. Considering heterogeneities by gender and age, the directions of early disaster math estimates are the same in 20 out of 24 cases between Tables 8 and D.8, and in 23 out of 24 cases between Tables 8 and E.8. In cases where directions differ, estimates are not significant.

Third, in terms of estimates of recent life disaster enrollment effects, in Tables 5, D.6 and E.6, we also find similar magnitudes and directions of average effects. Considering heterogeneities by gender and age, the directions of recent disaster enrollment estimates in Tables 7 and D.8 are identical for 20 out of 24 estimates, and identical for 21 out of 24 estimates in Tables 7 and E.8. In cases where directions differ, estimates are not significant. For recent life disaster math score effects, while the estimates are imprecise, the weighted results from Tables D.7 and E.7 show significant negative effects while or unweighted estimate is close to zero. We discuss these results in the discussion of the main results. The fact that weights change these results significantly indicates there might be substantial effects heterogeneity for recent disaster exposures that changes the average effects under alternative sample weights.

Finally, in terms of the estimates for mid-child life disaster test score effects. Among all unweighted and weighted results, we find insignificant effects for all disasters and floods both overall and among gender and age subgroups. The overall effects for severe disasters and severe floods are weakly positively significant in Table E.7, and insignificant in Tables 6 and D.7. The age and gender breakdown of the effects of mid-child life severe disasters and severe floods exposures are similar in direction and magnitudes in Table 9, D.9, and E.9.

D Additional Figures and Tables (Online): Applying Sample Weight
*(fsweight/sum of fsweight*pop size)*

Table D.1: Summary statistics for all children

	Mean	SD	Min	Max	N
Panel A: Enrollment, math-test scores, attainment					
Ever enrolled (dummy)	0.89	0.31	0	1	140,750
Enrollment in last school year t-1 (dummy)	0.76	0.43	0	1	140,724
Enrollment in this school year t (dummy)	0.80	0.40	0	1	140,735
Have math Score (dummy)	0.92	0.27	0	1	85,603
Math score (total)	14.67	7.25	0	21	78,695
Attainment (highest)	3.33	3.20	0	16	140,695
Attainment at start of last school year t-1	2.71	2.94	0	16	140,696
Attainment at start of this school year t	3.30	3.19	0	16	140,695
Panel B: Child and parental characteristics					
Age of child	10.57	3.61	5	17	140,788
Female (dummy)	0.49	0.50	0	1	140,788
Mother ever educated (dummy)	0.60	0.49	0	1	140,726
Mother has secondary sch education (dummy)	0.31	0.46	0	1	140,726
Mother is living in same HH (dummy)	0.91	0.29	0	1	140,546
Father is living in same HH (dummy)	0.81	0.39	0	1	140,398

Note: This table is Table 2 with sample weight applied. For sample weight, here we use variable generated from *(fsweight/sum of fsweight*pop size)*.

Table D.2: Summary statistics of disaster exposures for all children

	Mean	SD	Min	Max
Panel A: Any disasters				
<i>Had disaster</i>				
in most recent 12 mo.	0.52	0.50	0	1
in year prior to 12 mo. ago	0.63	0.48	0	1
<i># of mo. with disaster</i>				
in mid-child life	10.40	11.11	0	60
in the first 1,000 days (early life)	3.26	3.97	0	24
Panel B: Floods				
<i>Had disaster</i>				
in most recent 12 mo.	0.29	0.46	0	1
in year prior to 12 mo. ago	0.54	0.50	0	1
<i># of mo. with disaster</i>				
in mid-child life	3.52	4.39	0	23
in the first 1,000 days (early life)	1.44	2.21	0	11
Panel C: Severe disasters				
<i>Had disaster</i>				
in most recent 12 mo.	0.24	0.43	0	1
in year prior to 12 mo. ago	0.34	0.48	0	1
<i># of mo. with disaster</i>				
in mid-child life	4.01	4.72	0	33
in the first 1,000 days (early life)	1.60	2.52	0	24
Panel D: Severe floods				
<i>Had disaster</i>				
in most recent 12 mo.	0.19	0.40	0	1
in year prior to 12 mo. ago	0.27	0.44	0	1
<i># of mo. with disaster</i>				
in mid-child life	2.36	2.94	0	20
in the first 1,000 days (early life)	0.97	1.82	0	11

Note: This table is Table 3 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.3: Summary statistics for educational outcomes by country

	Mean	SD	Min	Max	N
Bangladesh					
Enrollment in last school year t-1	0.78	0.41	0	1	39,386
Enrollment in this school year t	0.84	0.37	0	1	39,386
Attainment (highest)	3.87	3.17	0	14	39,385
Math score (total)	16.15	6.01	0	21	22,354
Kyrgyz Republic					
Enrollment in last school year t-1	0.90	0.31	0	1	3,889
Enrollment in this school year t	0.94	0.25	0	1	3,889
Attainment (highest)	3.95	3.31	0	13	3,889
Math score (total)	19.40	2.75	0	21	2,349
Mongolia					
Enrollment in last school year t-1	0.94	0.23	0	1	7,437
Enrollment in this school year t	0.94	0.24	0	1	7,437
Attainment (highest)	4.23	3.54	0	16	7,437
Math score (total)	19.07	3.53	0	21	4,546
Nepal					
Enrollment in last school year t-1	0.90	0.30	0	1	7,790
Enrollment in this school year t	0.90	0.30	0	1	7,790
Attainment (highest)	4.22	3.37	0	12	7,789
Math score (total)	16.09	6.54	0	21	4,617
Pakistan					
Enrollment in last school year t-1	0.64	0.48	0	1	69,017
Enrollment in this school year t	0.69	0.46	0	1	69,027
Attainment (highest)	2.76	3.28	0	13	68,999
Math score (total)	10.17	7.42	0	21	35,997
Thailand					
Enrollment in last school year t-1	0.98	0.14	0	1	9,475
Enrollment in this school year t	0.99	0.11	0	1	9,476
Attainment (highest)	3.36	2.66	0	9	9,466
Math score (total)	19.94	2.69	0	21	6,704
Turkmenistan					
Enrollment in last school year t-1	0.87	0.34	0	1	3,730
Enrollment in this school year t	0.92	0.28	0	1	3,730
Attainment (highest)	4.01	3.19	0	12	3,730
Math score (total)	20.11	2.03	0	21	2,128

Note: This table is Table C.1 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.4: Summary statistics for child and parent attributes by country

	Mean	SD	N
Bangladesh			
Age of child	11.02	3.69	39,386
Female	0.49	0.50	39,386
Mother ever educated	0.73	0.45	39,386
Mother has secondary sch. education	0.44	0.50	39,386
Mother is living in same HH	0.93	0.26	39,374
Father is living in same HH	0.83	0.38	39,352
Kyrgyz Republic			
Age of child	10.34	3.59	3,889
Female	0.47	0.50	3,889
Mother ever educated	0.99	0.09	3,889
Mother has secondary sch. education	0.90	0.29	3,889
Mother is living in same HH	0.86	0.35	3,880
Father is living in same HH	0.78	0.41	3,871
Mongolia			
Age of child	10.10	3.66	7,437
Female	0.49	0.50	7,437
Mother ever educated	0.94	0.23	7,410
Mother has secondary sch. education	0.68	0.47	7,410
Mother is living in same HH	0.87	0.34	7,432
Father is living in same HH	0.75	0.43	7,345
Nepal			
Age of child	10.99	3.68	7,790
Female	0.49	0.50	7,790
Mother ever educated	0.50	0.50	7,790
Mother has secondary sch. education	0.22	0.41	7,790
Mother is living in same HH	0.91	0.29	7,787
Father is living in same HH	0.69	0.46	7,780
Pakistan			
Age of child	10.56	3.72	69,078
Female	0.48	0.50	69,078
Mother ever educated	0.36	0.48	69,048
Mother has secondary sch. education	0.15	0.36	69,048
Mother is living in same HH	0.96	0.19	68,906
Father is living in same HH	0.88	0.32	68,980
Thailand			
Age of child	9.66	2.88	9,477
Female	0.47	0.50	9,477
Mother ever educated	0.97	0.18	9,472
Mother has secondary sch. education	0.44	0.50	9,472
Mother is living in same HH	0.74	0.44	9,442
Father is living in same HH	0.64	0.48	9,352
Turkmenistan			
Age of child	10.03	3.64	3,731
Female	0.49	0.50	3,731
Mother ever educated	1	1	3,731
Mother has secondary sch. education	0.22	0.41	3,731
Mother is living in same HH	0.95	0.22	3,725
Father is living in same HH	0.88	0.33	3,718

Note: This table is Table C.2 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.5: Summary statistics for disaster exposure by country

	Mean	SD	Min	Max	N
Bangladesh					
<i>Had any disaster</i>					
in most recent 12 mo.	0.67	0.47	0	1	39,386
in year prior to 12 mo. ago	0.68	0.47	0	1	39,386
<i># of mo. with any disaster</i>					
in mid-child life	6.83	7.46	0	41	39,386
in the first 1,000 days (early life)	3.86	3.77	0	18	39,386
Kyrgyz Republic					
<i>Had any disaster</i>					
in most recent 12 mo.	0	0	0	0	3,889
in year prior to 12 mo. ago	0	0	0	0	3,889
<i># of mo. with any disaster</i>					
in mid-child life	3.15	3.22	0	13	3,889
in the first 1,000 days (early life)	1.50	1.63	0	5	3,889
Mongolia					
<i>Had any disaster</i>					
in most recent 12 mo.	0.93	0.26	0	1	7,437
in year prior to 12 mo. ago	0.93	0.26	0	1	7,437
<i># of mo. with any disaster</i>					
in mid-child life	5.85	5.35	0	21	7,437
in the first 1,000 days (early life)	2.80	3.12	0	10	7,437
Nepal					
<i>Had any disaster</i>					
in most recent 12 mo.	0.17	0.37	0	1	7,790
in year prior to 12 mo. ago	0.59	0.49	0	1	7,790
<i># of mo. with any disaster</i>					
in mid-child life	5.94	5.52	0	13	7,790
in the first 1,000 days (early life)	2.42	4.12	0	13	7,790
Pakistan					
<i>Had any disaster</i>					
in most recent 12 mo.	0.67	0.47	0	1	69,078
in year prior to 12 mo. ago	0.64	0.48	0	1	69,078
<i># of mo. with any disaster</i>					
in mid-child life	7.18	6.49	0	29	69,078
in the first 1,000 days (early life)	2.89	3.34	0	13	69,078
Thailand					
<i>Had any disaster</i>					
in most recent 12 mo.	0.04	0.18	0	1	9,477
in year prior to 12 mo. ago	0.61	0.49	0	1	9,477
<i># of mo. with any disaster</i>					
in mid-child life	28.76	10.66	0	60	9,477
in the first 1,000 days (early life)	3.74	5.50	0	24	9,477

Note: This table is Table C.4 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.6: Effects of disasters on enrollments

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
Had disaster in most recent 12 mo.	−0.008 (0.006)	−0.013** (0.006)	−0.001 (0.008)	−0.006 (0.007)
# of mo. with disaster first 1,000 days	−0.000 (0.001)	−0.002*** (0.001)	−0.002*** (0.001)	−0.003*** (0.001)
Female	−0.003 (0.002)	−0.003 (0.002)	−0.003 (0.002)	−0.003 (0.002)
Mother is alive	0.000 (0.008)	0.000 (0.008)	0.000 (0.008)	0.001 (0.008)
Father is alive	0.011** (0.004)	0.011** (0.004)	0.011** (0.004)	0.011** (0.004)
Mother is alive × living in same HH	0.021*** (0.006)	0.021*** (0.006)	0.021*** (0.006)	0.021*** (0.006)
Father is alive × living in same HH	−0.005** (0.003)	−0.005** (0.003)	−0.005** (0.003)	−0.005** (0.003)
Mother ever educated	0.037*** (0.003)	0.037*** (0.003)	0.037*** (0.003)	0.037*** (0.003)
Mother ever educated × secondary educ.	0.010*** (0.004)	0.010*** (0.004)	0.010*** (0.004)	0.010*** (0.004)
Observations	140,073	140,073	140,073	140,073
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Enrollment $t - 1$ × age group FE	Y	Y	Y	Y
Attainment t × age group FE	Y	Y	Y	Y
Enrollment $t - 1$ × country FE	Y	Y	Y	Y
Attainment t × country FE	Y	Y	Y	Y

Note: This table is Table 5 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.7: Effects of disasters on math scores

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
<i>Recent experience: had disaster</i>				
in most recent 12 mo.	-0.134 (0.200)	-0.125 (0.308)	-0.060 (0.239)	-0.402 (0.272)
in yr. prior 12 mo. ago	-0.391** (0.173)	-0.095 (0.274)	0.076 (0.167)	-0.100 (0.166)
<i>Mid-child life experience: # of mo. with disaster</i>				
Between early life & recent yr.	0.003 (0.018)	0.013 (0.032)	0.027 (0.024)	0.057 (0.037)
<i>Early-life experience: # of mo. with disaster</i>				
in the first 1,000 days	-0.015 (0.019)	-0.054** (0.027)	0.020 (0.027)	-0.024 (0.035)
Female	-0.413*** (0.062)	-0.413*** (0.063)	-0.413*** (0.062)	-0.415*** (0.063)
Mother is alive	0.563*** (0.182)	0.561*** (0.183)	0.557*** (0.182)	0.556*** (0.183)
Father is alive	0.181 (0.154)	0.176 (0.154)	0.180 (0.153)	0.179 (0.153)
Mother is alive $\times$ living in same HH	-0.204** (0.090)	-0.205** (0.091)	-0.202** (0.091)	-0.201** (0.091)
Father is alive $\times$ living in same HH	-0.198** (0.079)	-0.196** (0.079)	-0.197** (0.079)	-0.197** (0.079)
Mother ever educated	1.441*** (0.106)	1.438*** (0.106)	1.441*** (0.106)	1.438*** (0.106)
Mother ever educated $\times$ secondary education	0.963*** (0.129)	0.965*** (0.129)	0.966*** (0.129)	0.962*** (0.129)
Observations	78,296	78,296	78,296	78,296
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Attainment $t \times$ country FE	Y	Y	Y	Y

Note: This table is Table 6 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.8: Disasters and enrollments, heterogeneity across gender and age groups

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
Had disaster in most recent 12 mo.				
× Male				
× Age 5–8	−0.001 (0.009)	−0.017* (0.010)	0.013 (0.011)	0.037*** (0.011)
× Age 9–12	−0.009 (0.007)	−0.020** (0.008)	−0.018** (0.009)	−0.039*** (0.008)
× Age 13–17	−0.011 (0.010)	−0.010 (0.009)	−0.015 (0.010)	−0.041*** (0.008)
× Female				
× Age 5–8	−0.015 (0.011)	−0.024** (0.012)	0.011 (0.012)	0.045*** (0.010)
× Age 9–12	−0.008 (0.007)	−0.001 (0.007)	0.007 (0.009)	−0.005 (0.007)
× Age 13–17	−0.004 (0.009)	−0.004 (0.008)	−0.002 (0.009)	−0.022** (0.009)
# of mo. with disaster in the first 1,000 days				
× Male				
× Age 5–8	0.001 (0.001)	−0.001 (0.003)	−0.000 (0.002)	−0.003 (0.003)
× Age 9–12	−0.002*** (0.001)	−0.002 (0.001)	−0.002** (0.001)	−0.003 (0.002)
× Age 13–17	−0.001 (0.001)	−0.002* (0.001)	−0.003*** (0.001)	−0.001 (0.001)
× Female				
× Age 5–8	0.001 (0.001)	−0.007*** (0.003)	−0.003 (0.002)	−0.011*** (0.003)
× Age 9–12	0.000 (0.001)	−0.002 (0.001)	0.000 (0.001)	−0.001 (0.002)
× Age 13–17	−0.001 (0.001)	−0.002 (0.001)	−0.003* (0.001)	−0.002 (0.002)
Observations	140,073	140,073	140,073	140,073
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Enrollment $t - 1$ × age group FE	Y	Y	Y	Y
Attainment t × age group FE	Y	Y	Y	Y
Enrollment $t - 1$ × country FE	Y	Y	Y	Y
Attainment t × country FE	Y	Y	Y	Y

Note: This table is Table 7 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

Table D.9: Disasters and math scores, heterogeneity across gender and age groups

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
# of mo. with disaster in mid-child life				
× Male				
× Age 7–9	−0.019 (0.026)	0.027 (0.054)	0.010 (0.047)	0.083 (0.064)
× Age 10–12	−0.009 (0.025)	0.017 (0.050)	0.013 (0.030)	0.084 (0.056)
× Age 13–14	−0.007 (0.024)	−0.006 (0.044)	−0.011 (0.027)	0.056 (0.053)
× Female				
× Age 7–9	−0.002 (0.029)	0.030 (0.054)	0.014 (0.038)	0.065 (0.063)
× Age 10–12	0.006 (0.024)	−0.061 (0.049)	−0.020 (0.030)	−0.008 (0.053)
× Age 13–14	0.013 (0.023)	0.003 (0.043)	0.004 (0.026)	0.062 (0.047)
# of mo. with disaster in the first 1,000 days				
× Male				
× Age 7–9	−0.033 (0.035)	−0.114* (0.064)	−0.039 (0.040)	−0.066 (0.068)
× Age 10–12	0.026 (0.024)	0.001 (0.034)	0.024 (0.032)	−0.020 (0.051)
× Age 13–14	−0.011 (0.032)	−0.028 (0.051)	0.005 (0.046)	−0.039 (0.061)
× Female				
× Age 7–9	−0.026 (0.032)	−0.051 (0.067)	0.011 (0.039)	−0.011 (0.071)
× Age 10–12	0.010 (0.023)	−0.014 (0.036)	0.021 (0.028)	0.030 (0.049)
× Age 13–14	−0.035 (0.028)	−0.064 (0.041)	−0.012 (0.037)	−0.062 (0.050)
Observations	78,294	78,294	78,294	78,294
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Attainment $t \times$ country FE	Y	Y	Y	Y

Note: This table is Table 9 with sample weight applied. For sample weight, here we use variable generated from $(fsweight/sum\ of\ fsweight*pop\ size)$.

E Additional Figures and Tables (Online): Applying Sample Weight (*fsweight*)

Table E.1: Summary statistics for all children

	Mean	SD	Min	Max	N
Panel A: Enrollment, math-test scores, attainment					
Ever enrolled (dummy)	0.85	0.35	0	1	140,750
Enrollment in last school year t-1 (dummy)	0.72	0.45	0	1	140,724
Enrollment in this school year t (dummy)	0.76	0.43	0	1	140,735
Have math score (dummy)	0.92	0.28	0	1	85,603
Math score (total)	13.01	7.57	0	21	78,695
Attainment (highest)	3.18	3.29	0	16	140,695
Attainment at start of last school year t-1	2.59	3	0	16	140,696
Attainment at start of this school year t	3.14	3.27	0	16	140,695
Panel B: Child and parental characteristics					
Age of child	10.59	3.69	5	17	140,788
Female (dummy)	0.49	0.50	0	1	140,788
Mother ever educated (dummy)	0.51	0.50	0	1	140,726
Mother has secondary-school education (dummy)	0.26	0.44	0	1	140,726
Mother is living in same household (dummy)	0.94	0.25	0	1	140,546
Father is living in same household (dummy)	0.84	0.37	0	1	140,398

Note: This table is Table 2 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.2: Summary statistics of disaster exposures for all children

	Mean	SD	Min	Max
Panel A: Any disasters				
<i>Had disaster</i>				
in most recent 12 mo.	0.54	0.50	0	1
in year prior to 12 mo. ago	0.63	0.48	0	1
<i># of mo. with disaster</i>				
in mid-child life	7.10	8.32	0	60
in the first 1,000 days (early life)	2.84	3.52	0	24
Panel B: Floods				
<i>Had disaster</i>				
in most recent 12 mo.	0.28	0.45	0	1
in year prior to 12 mo. ago	0.52	0.50	0	1
<i># of mo. with disaster</i>				
in mid-child life	3.69	4.58	0	23
in the first 1,000 days (early life)	1.47	2.17	0	11
Panel C: Severe disasters				
<i>Had disaster</i>				
in most recent 12 mo.	0.31	0.46	0	1
in year prior to 12 mo. ago	0.31	0.46	0	1
<i># of mo. with disaster</i>				
in mid-child life	3.77	4.38	0	33
in the first 1,000 days (early life)	1.35	2.21	0	24
Panel D: Severe floods				
<i>Had disaster</i>				
in most recent 12 mo.	0.16	0.37	0	1
in year prior to 12 mo. ago	0.13	0.34	0	1
<i># of mo. with disaster</i>				
in mid-child life	2.29	2.91	0	20
in the first 1,000 days (early life)	0.88	1.66	0	11

Note: This table is Table 3 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.3: Summary statistics for educational outcomes by country

	Mean	SD	Min	Max	N
Bangladesh					
Enrollment in last school year t-1	0.78	0.41	0	1	39,386
Enrollment in this school year t	0.84	0.37	0	1	39,386
Attainment (highest)	3.87	3.17	0	14	39,385
Math score (total)	16.15	6.01	0	21	22,354
Kyrgyz Republic					
Enrollment in last school year t-1	0.90	0.31	0	1	3,889
Enrollment in this school year t	0.94	0.25	0	1	3,889
Attainment (highest)	3.95	3.31	0	13	3,889
Math score (total)	19.40	2.75	0	21	2,349
Mongolia					
Enrollment in last school year t-1	0.94	0.23	0	1	7,437
Enrollment in this school year t	0.94	0.24	0	1	7,437
Attainment (highest)	4.23	3.54	0	16	7,437
Math score (total)	19.07	3.53	0	21	4,546
Nepal					
Enrollment in last school year t-1	0.90	0.30	0	1	7,790
Enrollment in this school year t	0.90	0.30	0	1	7,790
Attainment (highest)	4.22	3.37	0	12	7,789
Math score (total)	16.09	6.54	0	21	4,617
Pakistan					
Enrollment in last school year t-1	0.64	0.48	0	1	69,017
Enrollment in this school year t	0.68	0.47	0	1	69,027
Attainment (highest)	2.74	3.26	0	13	68,999
Math score (total)	10.02	7.35	0	21	35,997
Thailand					
Enrollment in last school year t-1	0.98	0.14	0	1	9,475
Enrollment in this school year t	0.99	0.12	0	1	9,476
Attainment (highest)	3.17	2.60	0	9	9,466
Math score (total)	19.66	3.09	0	21	6,704
Turkmenistan					
Enrollment in last school year t-1	0.87	0.34	0	1	3,730
Enrollment in this school year t	0.92	0.28	0	1	3,730
Attainment (highest)	4.01	3.19	0	12	3,730
Math score (total)	20.11	2.03	0	21	2,128

Note: This table is Table C.1 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.4: Summary statistics for child and parent attributes by country

	Mean	SD	N
Bangladesh			
Age of child	11.02	3.69	39,386
Female	0.49	0.50	39,386
Mother ever educated	0.73	0.45	39,386
Mother has secondary sch. education	0.44	0.50	39,386
Mother is living in same HH	0.93	0.26	39,374
Father is living in same HH	0.83	0.38	39,352
Kyrgyz Republic			
Age of child	10.34	3.59	3,889
Female	0.47	0.50	3,889
Mother ever educated	0.99	0.09	3,889
Mother has secondary sch. education	0.90	0.29	3,889
Mother is living in same HH	0.86	0.35	3,880
Father is living in same HH	0.78	0.41	3,871
Mongolia			
Age of child	10.10	3.66	7,437
Female	0.49	0.50	7,437
Mother ever educated	0.94	0.23	7,410
Mother has secondary sch. education	0.68	0.47	7,410
Mother is living in same HH	0.87	0.34	7,432
Father is living in same HH	0.75	0.43	7,345
Nepal			
Age of child	10.99	3.68	7,790
Female	0.49	0.50	7,790
Mother ever educated	0.50	0.50	7,790
Mother has secondary sch. education	0.22	0.41	7,790
Mother is living in same HH	0.91	0.29	7,787
Father is living in same HH	0.69	0.46	7,780
Pakistan			
Age of child	10.56	3.72	69,078
Female	0.48	0.50	69,078
Mother ever educated	0.34	0.47	69,048
Mother has secondary sch. education	0.14	0.35	69,048
Mother is living in same HH	0.96	0.19	68,906
Father is living in same HH	0.88	0.33	68,980
Thailand			
Age of child	9.49	2.83	9,477
Female	0.48	0.50	9,477
Mother ever educated	0.95	0.22	9,472
Mother has secondary sch. education	0.35	0.48	9,472
Mother is living in same HH	0.69	0.46	9,442
Father is living in same HH	0.58	0.49	9,352
Turkmenistan			
Age of child	10.03	3.64	3,731
Female	0.49	0.50	3,731
Mother ever educated	1	1	3,731
Mother has secondary sch. education	0.22	0.41	3,731
Mother is living in same HH	0.95	0.22	3,725
Father is living in same HH	0.88	0.33	3,718

Note: This table is Table C.2 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.5: Summary statistics for disaster exposure by country

	Mean	SD	Min	Max	N
Bangladesh					
<i>Had any disaster</i>					
in most recent 12 mo.	0.67	0.47	0	1	39,386
in year prior to 12 mo. ago	0.68	0.47	0	1	39,386
<i># of mo. with any disaster</i>					
in mid-child life	6.83	7.46	0	41	39,386
in the first 1,000 days (early life)	3.86	3.77	0	18	39,386
Kyrgyz Republic					
<i>Had any disaster</i>					
in most recent 12 mo.	0	0	0	0	3,889
in year prior to 12 mo. ago	0	0	0	0	3,889
<i># of mo. with any disaster</i>					
in mid-child life	3.15	3.22	0	13	3,889
in the first 1,000 days (early life)	1.50	1.63	0	5	3,889
Mongolia					
<i>Had any disaster</i>					
in most recent 12 mo.	0.93	0.26	0	1	7,437
in year prior to 12 mo. ago	0.93	0.26	0	1	7,437
<i># of mo. with any disaster</i>					
in mid-child life	5.85	5.35	0	21	7,437
in the first 1,000 days (early life)	2.80	3.12	0	10	7,437
Nepal					
<i>Had any disaster</i>					
in most recent 12 mo.	0.17	0.37	0	1	7,790
in year prior to 12 mo. ago	0.59	0.49	0	1	7,790
<i># of mo. with any disaster</i>					
in mid-child life	5.94	5.52	0	13	7,790
in the first 1,000 days (early life)	2.42	4.12	0	13	7,790
Pakistan					
<i>Had any disaster</i>					
in most recent 12 mo.	0.58	0.49	0	1	69,078
in year prior to 12 mo. ago	0.65	0.48	0	1	69,078
<i># of mo. with any disaster</i>					
in mid-child life	6.29	6.38	0	29	69,078
in the first 1,000 days (early life)	2.57	3.19	0	13	69,078
Thailand					
<i>Had any disaster</i>					
in most recent 12 mo.	0.06	0.24	0	1	9,477
in year prior to 12 mo. ago	0.52	0.50	0	1	9,477
<i># of mo. with any disaster</i>					
in mid-child life	28.75	13.35	0	60	9,477
in the first 1,000 days (early life)	4.60	5.51	0	24	9,477

Note: This table is Table C.4 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.6: Effects of disasters on enrollments

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
Had disaster in most recent 12 mo.	-0.011* (0.007)	-0.016*** (0.006)	0.003 (0.009)	-0.004 (0.008)
# of mo. with disaster first 1,000 days	-0.001** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.003*** (0.001)
Female	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)
Mother is alive	-0.003 (0.008)	-0.003 (0.008)	-0.003 (0.008)	-0.003 (0.008)
Father is alive	0.015*** (0.005)	0.015*** (0.005)	0.015*** (0.005)	0.015*** (0.005)
Mother is alive $\times$ living in same HH	0.024*** (0.005)	0.024*** (0.005)	0.024*** (0.005)	0.024*** (0.005)
Father is alive $\times$ living in same HH	-0.008*** (0.003)	-0.008*** (0.003)	-0.008*** (0.003)	-0.008*** (0.003)
Mother ever educated	0.039*** (0.003)	0.039*** (0.003)	0.039*** (0.003)	0.039*** (0.003)
Mother ever educated $\times$ secondary educ.	0.013*** (0.003)	0.013*** (0.003)	0.013*** (0.002)	0.013*** (0.003)
Observations	140,073	140,073	140,073	140,073
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Enrollment $t - 1 \times$ age group FE	Y	Y	Y	Y
Attainment $t \times$ age group FE	Y	Y	Y	Y
Enrollment $t - 1 \times$ country FE	Y	Y	Y	Y
Attainment $t \times$ country FE	Y	Y	Y	Y

Note: This table is Table 5 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.7: Effects of disasters on math scores

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
<i>Recent experience: had disaster</i>				
in most recent 12 mo.	−0.108 (0.216)	−0.063 (0.363)	−0.086 (0.285)	−0.450* (0.263)
in yr. prior 12 mo. ago	−0.413* (0.237)	−0.077 (0.340)	0.361 (0.289)	−0.047 (0.196)
<i>Mid-child life experience: # of mo. with disaster</i>				
Between early life & recent yr.	0.009 (0.019)	0.023 (0.035)	0.043* (0.025)	0.074* (0.042)
<i>Early-life experience: # of mo. with disaster</i>				
in the first 1,000 days	−0.027 (0.018)	−0.067** (0.027)	0.004 (0.027)	−0.040 (0.035)
Female	−0.602*** (0.080)	−0.602*** (0.080)	−0.601*** (0.080)	−0.604*** (0.080)
Mother is alive	0.331* (0.194)	0.326* (0.194)	0.328* (0.194)	0.323* (0.194)
Father is alive	0.242 (0.155)	0.237 (0.155)	0.240 (0.155)	0.240 (0.155)
Mother is alive × living in same HH	−0.049 (0.102)	−0.048 (0.102)	−0.047 (0.102)	−0.046 (0.101)
Father is alive × living in same HH	−0.177** (0.085)	−0.175** (0.085)	−0.175** (0.085)	−0.176** (0.084)
Mother ever educated	1.429*** (0.095)	1.425*** (0.094)	1.429*** (0.095)	1.426*** (0.094)
Mother ever educated × secondary education	1.039*** (0.078)	1.039*** (0.079)	1.041*** (0.079)	1.036*** (0.079)
Observations	78,296	78,296	78,296	78,296
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Attainment $t \times$ country FE	Y	Y	Y	Y

Note: This table is Table 6 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.8: Disasters and enrollments, heterogeneity across gender and age groups

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
Had disaster in most recent 12 mo.				
× Male				
× Age 5–8	−0.001 (0.011)	−0.031*** (0.011)	0.008 (0.013)	0.041*** (0.011)
× Age 9–12	−0.015* (0.008)	−0.020** (0.009)	−0.009 (0.011)	−0.035*** (0.009)
× Age 13–17	−0.025** (0.012)	−0.012 (0.010)	−0.016 (0.011)	−0.050*** (0.008)
× Female				
× Age 5–8	−0.003 (0.012)	−0.031** (0.014)	0.020 (0.014)	0.059*** (0.011)
× Age 9–12	−0.013* (0.007)	−0.006 (0.007)	0.011 (0.011)	−0.002 (0.007)
× Age 13–17	−0.011 (0.010)	−0.003 (0.009)	0.003 (0.010)	−0.020** (0.009)
# of mo. with disaster in the first 1,000 days				
× Male				
× Age 5–8	0.001 (0.001)	0.003 (0.003)	0.001 (0.002)	−0.001 (0.003)
× Age 9–12	−0.003*** (0.001)	−0.003** (0.001)	−0.004*** (0.001)	−0.004* (0.002)
× Age 13–17	−0.001 (0.001)	−0.003** (0.001)	−0.003** (0.001)	−0.001 (0.001)
× Female				
× Age 5–8	−0.000 (0.001)	−0.005 (0.003)	−0.004* (0.002)	−0.009*** (0.003)
× Age 9–12	−0.000 (0.001)	−0.002** (0.001)	−0.000 (0.001)	−0.001 (0.002)
× Age 13–17	−0.001 (0.001)	−0.002* (0.001)	−0.002 (0.001)	−0.002 (0.002)
Observations	140,073	140,073	140,073	140,073
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Enrollment $t - 1$ × age group FE	Y	Y	Y	Y
Attainment t × age group FE	Y	Y	Y	Y
Enrollment $t - 1$ × country FE	Y	Y	Y	Y
Attainment t × country FE	Y	Y	Y	Y

Note: This table is Table 7 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.

Table E.9: Disasters and math scores, heterogeneity across gender and age groups

	(1) All disasters	(2) Floods	(3) Severe disasters	(4) Severe floods
# of mo. with disaster in mid-child life				
× Male				
× Age 7–9	−0.016 (0.029)	−0.020 (0.061)	0.011 (0.052)	0.043 (0.073)
× Age 10–12	−0.003 (0.026)	−0.011 (0.058)	0.023 (0.033)	0.061 (0.063)
× Age 13–14	−0.011 (0.025)	−0.050 (0.052)	−0.007 (0.031)	0.019 (0.058)
× Female				
× Age 7–9	0.016 (0.029)	0.026 (0.061)	0.047 (0.042)	0.089 (0.073)
× Age 10–12	0.011 (0.026)	−0.080 (0.058)	−0.015 (0.031)	−0.021 (0.058)
× Age 13–14	0.017 (0.025)	−0.024 (0.050)	0.017 (0.029)	0.046 (0.051)
# of mo. with disaster in the first 1,000 days				
× Male				
× Age 7–9	−0.056 (0.039)	−0.165** (0.074)	−0.071* (0.041)	−0.120* (0.068)
× Age 10–12	0.019 (0.026)	−0.008 (0.038)	0.014 (0.035)	−0.026 (0.057)
× Age 13–14	−0.005 (0.047)	−0.029 (0.062)	−0.001 (0.055)	−0.011 (0.076)
× Female				
× Age 7–9	−0.061* (0.032)	−0.157** (0.073)	−0.042 (0.045)	−0.114* (0.067)
× Age 10–12	0.011 (0.024)	−0.017 (0.037)	0.043 (0.031)	0.014 (0.049)
× Age 13–14	−0.026 (0.032)	−0.027 (0.049)	0.011 (0.047)	−0.038 (0.063)
Observations	78,294	78,294	78,294	78,294
Within-country location FE	Y	Y	Y	Y
Interview year FE	Y	Y	Y	Y
Interview month FE	Y	Y	Y	Y
Child age FE	Y	Y	Y	Y
Attainment $t \times$ country FE	Y	Y	Y	Y

Note: This table is Table 9 with sample weight applied. For sample weight, here we use *fsweight* from MICS data.